



Time Series Analysis, Monte Carlo Simulation, and European Option Pricing of Apple Inc. Stock

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Project Presentation Agenda

Overview of our quantitative pricing methodology and investment strategy

- 1 Investment Challenge
- 2 Our Solution
- 3 Project Objectives
- 4 Data & Statistical Validation
- 5 Market Simulation (GBM)
- 6 Option Pricing Results
- 7 Quantitative Pricing Framework
- 8 Business Value & Investment Opportunity

Conclusion

Core Challenges

The Investment Challenge

Financial markets are uncertain, making investment decisions and option pricing difficult. Investors need reliable tools that can analyze historical market behavior, simulate possible future price movements, quantify investment risk, and value financial derivatives using a transparent quantitative framework.

- Market uncertainty
- Risk estimation
- Accurate option valuation



Our platform transforms historical market data into actionable investment intelligence.

Our FinTech Solution

A Python-based quantitative pricing platform that transforms historical market data into probabilistic future price scenarios, realistic market simulations, and accurate option valuations for informed investment decisions.

Time Series Analysis

Extracts trends, patterns, and statistical characteristics from historical financial data.

ARIMA Forecasting

Models short-term market dynamics and identifies underlying time-series behaviour.

Quantitative
Pricing
Engine

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graph TD; TSA[Time Series Analysis] --> QPE((Quantitative Pricing Engine)); ARIMA[ARIMA Forecasting] --> QPE; MCP[Monte Carlo Option Pricing] --> QPE; GBM[GBM Simulation] --> QPE;
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Monte Carlo Option Pricing

Values European options by simulating thousands of potential market scenarios.

GBM Simulation

Generates realistic future stock price paths using stochastic modelling.

One integrated platform for financial forecasting, simulation, and derivative pricing.

Project Objectives

A Roadmap to Quantitative Excellence



Data Foundation

Analyse historical AAPL data and investigate core statistical properties.



Stochastic Parameterization

Accurately estimate Geometric Brownian Motion (GBM) parameters.



Derivative Valuation

Price European call and put options using the simulation engine.



Time-Series Modeling

Identify the optimal ARIMA model for volatility capture.



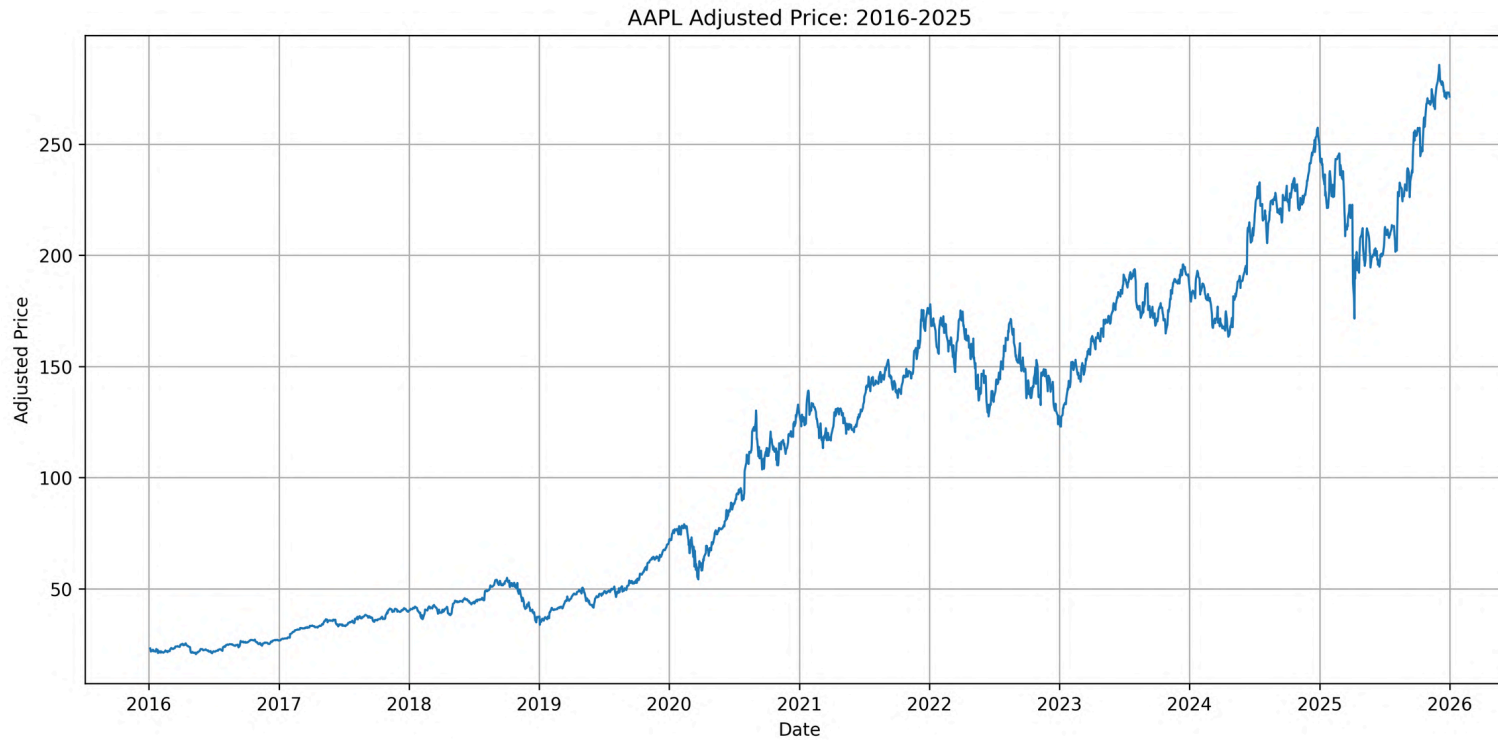
Future Simulation

Execute Monte Carlo simulations for future stock price trajectories.



Validation & Benchmarking

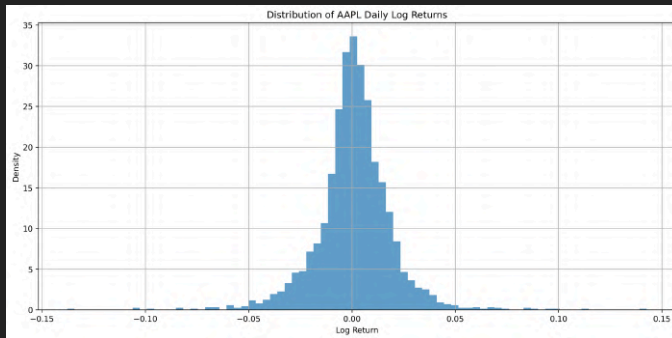
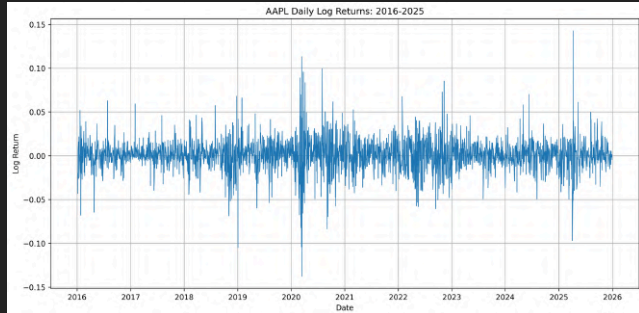
Validate model outputs against the theoretical Black-Scholes framework.



- Daily stock prices exhibit a non-stationary trend.
- Daily log returns fluctuate around zero.
- Rolling statistics indicate changing market volatility.
- Volatility clustering is present.
- Returns are suitable for statistical modelling.

Data & Statistical Validation

AAPL Quantitative Diagnostic Overview

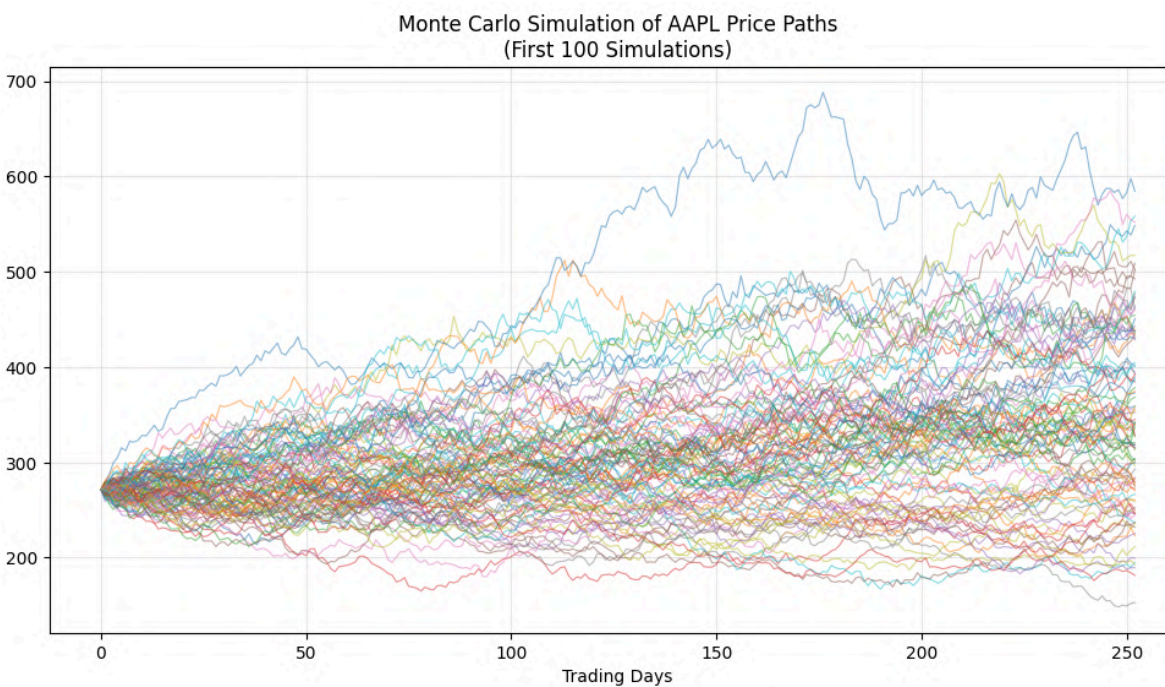


- Price Series: Non - Stationary
- Log Return: Stationary
- Best model: ARIMA (1, 0, 0)
- AIC: -12984.70
- BIC: -12967.21
- Annual Volatility: 29.02%
- Jarque-Bera: Non-normal Return

The statistical diagnostics provide confidence that the return series is suitable for forecasting and simulation. These results demonstrate that the return series is statistically suitable for time-series modelling and parameter estimation.

Simulation using Geometric Brownian Motion

Quantifying risk through 10,000 realistic market scenarios



Parameter	Values
Spot price	271.36
Mean	24.44%
Volatility	29.02%
Risk-free r	4.5%
Time Horizon T	1 Year
Simulation	10,000
Mean Simulated Price	345.67
Min. Simulated Price	121.73
Max. Simulated Price	974.91
Probability of Gain	74.48%
Probability of Loss	24.52%

BLACK-SCHOLES MODEL

European Call Option Price

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

where

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

S_0 = Current stock price

K = Strike price

r = Risk-free interest rate

σ = Volatility of the stock

T = Time to maturity

$N(\cdot)$ = Cumulative standard normal distribution

Performance Analysis

European Option Pricing

Monte Carlo simulation produces option prices that closely match the Black-Scholes benchmark, confirming the reliability of the pricing engine.

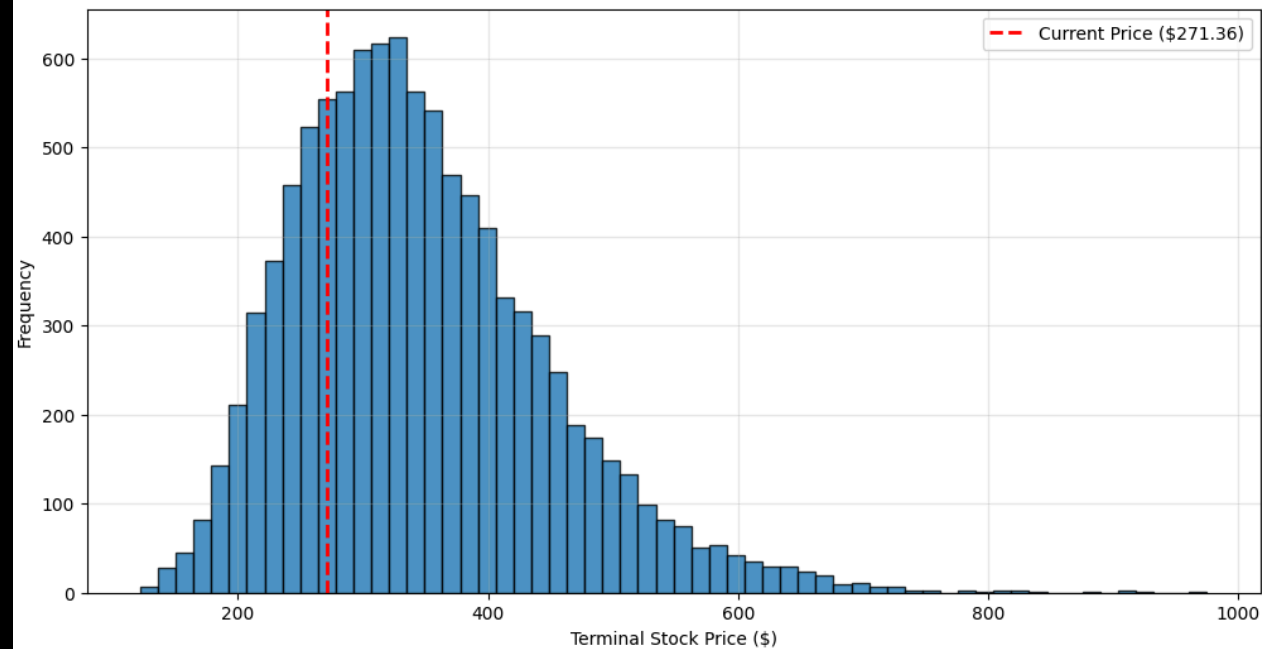
Comparison Table

European Call: 36.48 (Monte Carlo) vs 36.95 (Black-Scholes)
| Diff: 1.27%

European Put: 25.19 (Monte Carlo) vs 25.01 (Black-Scholes)
| Diff: 0.70%

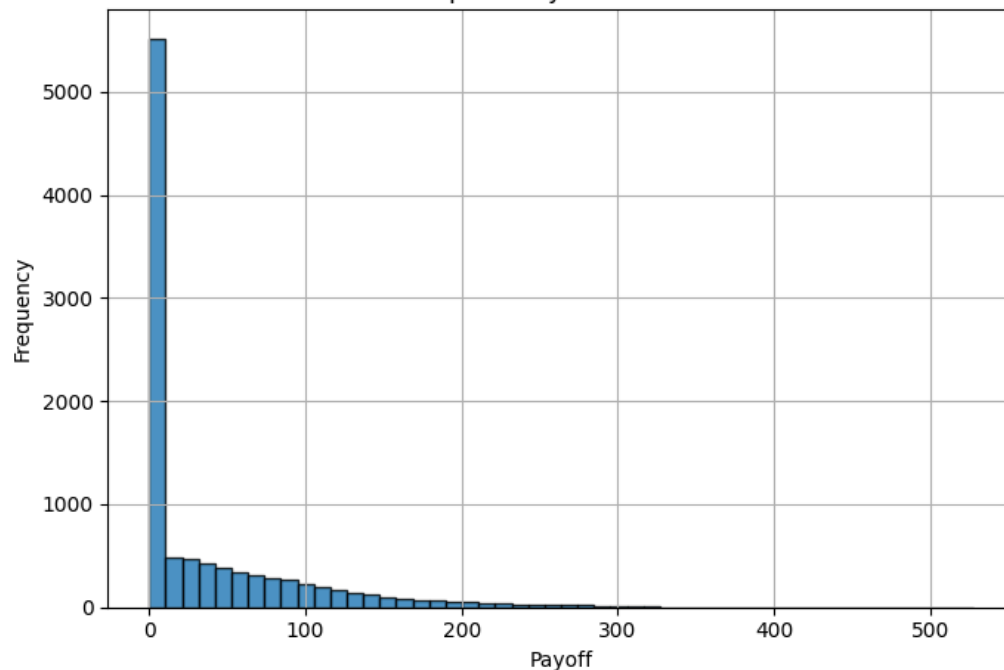
Agreement with Black-Scholes validates the pricing engine.

Distribution of Simulated AAPL Prices After One Year

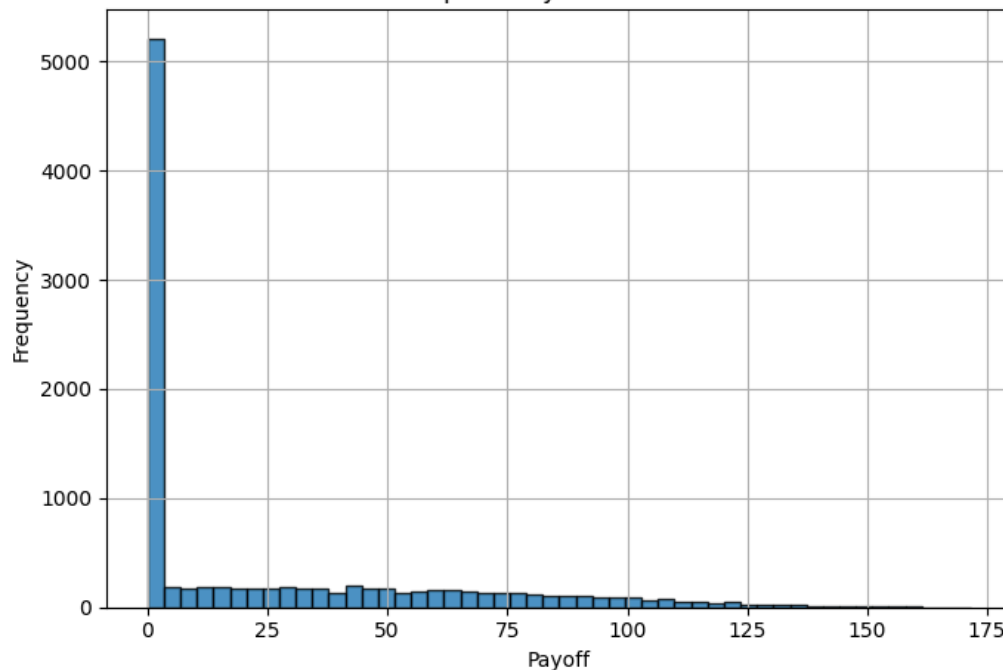


- Uses real historical market data.
- Statistically validated models.
- Simulates thousands of realistic future scenarios.
- Flexible pricing framework.
- Easily extendable to cryptocurrencies, commodities, FX, and indices.
- Built entirely in Python for transparency and reproducibility.

Call Option Payoff Distribution



Put Option Payoff Distribution



Quantitative Pricing Framework

A statistically validated workflow from raw market data to option valuation

- **Data Preparation & Analysis**

Historical AAPL returns were analysed using time-series techniques.

- **Modeling & Estimation**

ARIMA(1,0,0) captured limited short-term return dynamics.

- **Simulation Strategy**

GBM generated future price scenarios.

- **Valuation & Validation**

Monte Carlo prices are very close to the Black-Scholes benchmark. This validates our simulation framework.

THANK YOU

GBM assumes constant volatility and normally distributed returns.

Future extensions include GARCH, stochastic volatility, jump-diffusion models, and American option pricing.